

Effects of VR Visual Feedback on the Perceptual Characteristics of Tendon Vibration–Induced Kinesthetic Illusion

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ABSTRACT

Kinesthetic illusion is a promising technique for providing proprioceptive feedback in neurorehabilitation and virtual reality (VR). However, reliable methods for inducing strong illusions remain unclear. This study investigates the effect of VR-based visual feedback on kinesthetic illusion perception. A virtual right hand moving in accordance with the kinesthetic illusion is presented via a head-mounted display and compared with an eyes-closed condition. The results show that illusion perception is enhanced by synchronized virtual hand feedback, with statistically significant differences observed under specific experimental conditions, indicating that VR visual feedback effectively modulates the strength of kinesthetic illusions.

Index Terms: Kinesthetic illusion, flexor carpi radialis tendon, proprioception, HMD

1 INTRODUCTION

The phenomenon of kinesthetic illusion [1] has gained attention as a rehabilitation technique for providing proprioceptive feedback in neurorehabilitation and virtual reality (VR). Kinesthetic illusion is proprioceptive phenomena in which vibratory stimulation applied to tendons induces muscle spindle activation, generating a sensation of movement without actual motion. For rehabilitation applications, stable induction of kinesthetic illusion is essential. Although various methods have been proposed, including multi-point muscle stimulation [2] and tendon frequency–based stimulation [3], stable induction has not yet been achieved. Approaches combining visual stimuli have also been explored [4][5], but seamless integration remains challenging.

Therefore, this study aims to achieve seamless visual feedback by reproducing the kinesthetic illusion generated by vibration stimulation on the right wrist at the left wrist, then displaying the resulting movement of the left wrist as movement of the right wrist via VR goggles. This approach will verify the synergistic effect of integrating visual feedback with the kinesthetic illusion.

2 EXPERIMENT

The purpose of this study is to verify whether visual feedback via VR imagery contributes to enhancing the perception of kinesthetic illusion. Using the frequency of vibratory stimuli and the presence or absence of visual information as experimental parameters, we evaluated their effects on the maximum angle of the kinesthetic illusion.

2.1 Experimental setup

Figure 1 shows the configuration of the experimental apparatus used in this study. The apparatus consists of a “stimulus presentation unit” (right-hand side) that applies vibratory stimuli and an

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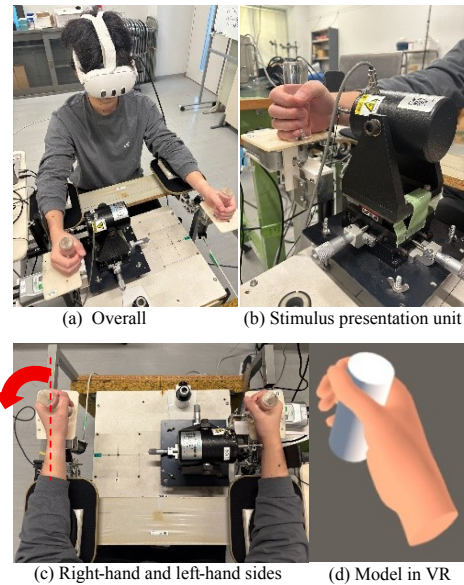


Figure 1: Vibration stimulus presentation device.

“evaluation unit” (left-hand side) that quantifies the magnitude of the resulting illusion. A function generator was used to generate the vibratory stimuli, enabling the output of arbitrary vibration patterns. To measure the illusion magnitude, the kinesthetic sensation perceived by the subject’s right hand was reproduced by the left hand’s motion, and its displacement was recorded using a rotary encoder. An HMD (Meta Quest 3) was used for image presentation. The virtual environment was constructed in Unity, and images were output from the PC to the HMD via Quest Link. Within the virtual environment, a virtual right hand displayed symmetrically synchronized movement with the left hand, visually superimposing the virtual motion onto the perceived kinesthetic illusion. Furthermore, to confirm the presence or absence of bodily responses unrelated to the illusion phenomenon—specifically the Tonic Vibration Reflex (TVR) and Antagonist Vibratory Response (AVR)—forces in the flexion and extension directions of the right hand were measured using a force gauge.

2.2 Experimental Method

In each trial, a sine-wave vibratory stimulus was applied to the flexor carpi radialis (FCR) muscle of the right wrist for 30 s, followed by a rest period of 30 s. Trials were randomly conducted under three visual conditions: eyes-closed (Condition 1), visual feedback presented via a head-mounted display (HMD) (Condition 2), and a combination of both conditions (Condition 3). In Condition 3, the visual feedback was switched during the vibration stimulation, such that Condition 1 was applied for the first 10 s, followed by Condition 2 for the subsequent 20 s.

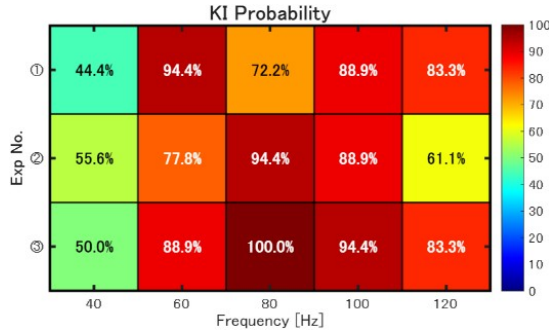


Figure 2: Probability of KI occurrence.

2.3 Experimental Conditions

The stimulus conditions consisted of five sine-wave vibration frequencies (40, 60, 80, 100, and 120 Hz), an acceleration amplitude of 100 m/s^2 , and an initial pressing force of 1.5 N. For each participant, six trials were conducted for each combination of visual condition and vibration frequency. The evaluation metric was the maximum joint angle of the kinesthetic illusion observed in each trial.

2.4 Experimental Results

First, we evaluated the occurrence probability of the kinesthetic illusion across all five frequencies. Figure 2 shows a heatmap of the occurrence probability calculated from the pooled data of all subjects, where an illusion was defined as a measured angle of 3° or greater. Since the probabilities were highest at 80 Hz and 100 Hz, subsequent analyses focused on these two frequencies. Because individual differences in sensitivity to visual stimuli may be substantial, we focused on within-subject comparisons as a first step to analyze the specific effects of visual conditions. Figure 3 shows the distribution of the maximum angle of the kinesthetic illusions for three subjects under these conditions. Overall, Condition 2 (HMD visual feedback) exhibited the highest mean and median values. Specifically, for Subject B at 80 Hz, an analysis of variance revealed a statistically significant difference among the conditions ($p < 0.05$). In contrast, for Subjects A and C, as well as under the 100 Hz condition, although Condition 2 tended to yield larger maximum angles compared to the other conditions, no statistically significant differences were observed. Furthermore, under the combined condition of eyes closed and HMD video presentation (Condition 3), the maximum angle was lower than that observed in Condition 2, or in some cases, remained at a level comparable to that of Condition 1.

3 DISCUSSION

The experimental results confirmed a tendency for the perceived magnitude of the kinesthetic illusions to increase when visual feedback via VR imagery was presented in addition to the vibration stimulus (Condition 2), compared to the eyes-closed condition (Condition 1). Statistically significant differences were observed for some subjects and frequency bands. These findings suggest the potential to artificially amplify and control the perceived magnitude of the kinesthetic illusion through the intervention of visual information. Conversely, in Condition 3, in which the video presentation was switched from the eyes-closed state midway through the trial, the enhancement effect tended to be less pronounced than in Condition 2, where the video was presented continuously throughout the stimulation period. This result indicates that the timing of visual stimulus presentation also influences the perception of the kinesthetic illusion. Furthermore, while a significant increase was observed in Subject B, no statistically significant differences were

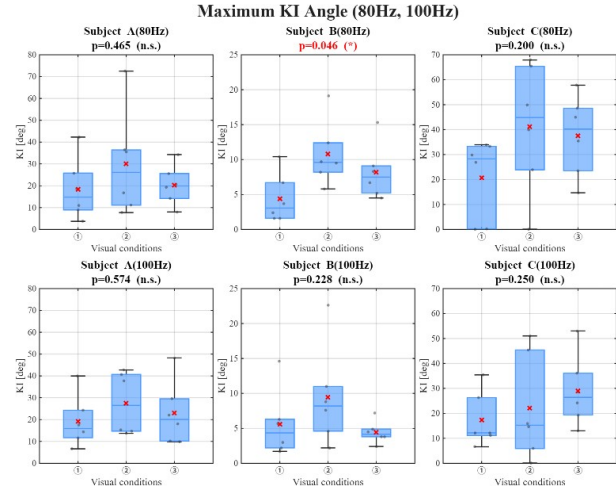


Figure 3: Maximum angle of kinesthetic illusions at 80, 100 Hz.

found for the other subjects. This suggests the presence of individual differences in the illusion-enhancing effect induced by visual stimuli.

4 CONCLUSION

The phenomenon of kinesthetic illusion has been attracting increasing attention as a technique for providing proprioceptive feedback in neuro-rehabilitation and virtual reality (VR) applications. However, methods for stably inducing strong kinesthetic illusions have not yet been fully established. In this study, we investigated the effect of visual feedback using VR on the perception of kinesthetic illusions. The experimental results indicate that, compared with the eyes-closed condition, presenting a virtual right hand that moves in accordance with the kinesthetic illusion via a head-mounted display (HMD) tends to enhance the perceived intensity of the illusion. Statistically significant differences were observed under specific experimental conditions. Future work will involve increasing the number of participants to further investigate the factors contributing to individual differences in illusion induction and to establish more effective and optimal visual presentation methods.

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